

Part 1: Master of the Dogma! Gimme Protein!!

1. First, test out your ability to use the central dogma to create a protein (chain of amino acids) starting from a DNA segment.
2. Use the sliders to line up the DNA and mRNA base sequences, and type in the mRNA bases.
3. Hint: In mRNA, the start codon is AUG. On the DNA template strand, this is TAC.
4. Open the amino acids codon chart in a new tab.
5. Transcribe from DNA TAC / mRNA AUG to the first in-frame stop codon. Click Check Transcription and correct any mistakes. Blue boxes will show the aligned DNA triplets and mRNA codons.
6. Next, use your codon chart to enter the 3-letter amino acid for each coding codon, up to the stop codon.
7. Again, ensure alignment using the sliders.
8. Once complete, click Check Amino Acids and correct where necessary.
9. Next, use the word codon chart (linked at the top) to translate the amino acids into words.
10. **NOTE:** The word system is made up to help build understanding: each word represents an amino acid, and the sentence represents the protein.
11. Then click Check Words and correct if necessary.

Part 2: Hey, don't mess with my Protein, you mutant!

- This part helps us understand the effect that a single change (mutation) to DNA can make.
- We represent an amino acid with a word.
- We therefore represent a protein with a sentence.
- In this model, a sentence that still makes sense represents a protein sequence that has been less disrupted.
- So how much can one change make??

Objectives:

- Investigate three types of mutation in this model: Deletion, Substitution and Insertion.
- Compare the original and mutated DNA, and use the blue codon boxes to follow the reading frame.
- What is the likely effect of each mutation type?
- Can you explain why each mutation has the effect that it does?
- What can cause a more severe effect?
- What can cause the effect to be minimal?
- Explain and evaluate the effect of each mutation type. As you work, make notes in your book.

Extension: Without words, make up your own DNA gene sequence on paper and try to create big and small effects for each mutation type.